Chapter 3 — Core ethical principles

This clip turns ethical slogans into operational tests

Learning objectives

- Name the five core principles, state what each demands in practice
- Recognize them as design constraints, not post-deployment patches

Five principles overview

- Non-maleficence asks teams to prevent privacy breaches, discrimination
- Fairness requires avoiding systematic disadvantage and monitoring disparities
- Transparency means disclosing collection purposes and
- Accountability assigns responsibility, monitoring, and remediation
- Respect for autonomy centers informed consent and ongoing control

Do no harm

- Do no harm requires anticipating harms from collection, processing
- Yielding less accurate predictions and misdiagnosis risk for that group
- Upholding non-maleficence means auditing subgroup performance and adding safeguards before

Example 3.2 — Healthcare AI system

- Example 3.2 — hands-on module
- The clinical pattern: skewed training data becomes skewed care
- Explore the chapter example module
- View files: `modules/chapter3/example2/`

Fairness

- Fairness requires that practices and models not systematically disadvantage particular
- The model can reproduce that pattern for new applicants
- Treat disparity risk as a design constraint, not a post-deployment discovery

Example 3.3 — Hiring algorithm

- Example 3.3 — hands-on module
- Example 3.3 restates the hiring pattern
- Chapter 7 develops metrics and mitigation
- Explore the chapter example module
- View files: `modules/chapter3/example3/`

Transparency and accountability

- Transparency builds trust by clarifying what is collected, why
- Accountability assigns someone who can explain outcomes, respond to incidents
- Example 3.4 applies transparency to credit scoring

Respect for autonomy

- Respect for autonomy centers informed consent and ongoing control
- Lets users revoke consent or delete history
- Autonomy fails when consent is buried in unrelated terms or withdrawal is practically

Example 3.6 — Health-tracking app

- Example 3.6 — hands-on module
- Example 3.6 pairs consent user experience with sensitive streams such as activity and
- Explore the chapter example module
- View files: `modules/chapter3/example6/`

Takeaways

- Principles are operational tests for design choices
- Non-maleficence and fairness catch harm before deployment
- Transparency and accountability make systems answerable
- Autonomy requires meaningful consent and control, not checkbox compliance alone

Next

- Complete the quiz for this part
- Continue to utility trade-offs, ethical dilemmas when principles conflict